

Normalized 3NF ERD - Condominium Billing and Forecasting System

Strict logical model: facts are stored once; calculated totals, balances, consumption, and payment status are produced from source relations.

Legend Core table Supporting table 3NF refinement — 1-to-many relation - - - - - optional/secondary relation

UserAccount	
Field	Key / rule
PK user_id : int	primary key
full_name : varchar	required
email : varchar	UK - unique login
password_hash : varchar	required
account_role : varchar	admin collector resident
account_status : varchar	active inactive
created_at : datetime	required

AuditLog	
Field	Key / rule
PK audit_id : int	primary key
FK actor_user_id : int	-> UserAccount
entity_name : varchar	audited table name
entity_id : int	record identifier
action : varchar	create update review
old_values : json	before change
new_values : json	after change
created_at : datetime	required

Unit	
Field	Key / rule
PK unit_id : int	primary key
unit_number : varchar	UK - unique
floor : varchar	required
association_dues_rate : decimal	current configured rate
occupancy_status : varchar	occupied vacant
created_at : datetime	required

UnitAssignment	
Field	Key / rule
PK assignment_id : int	primary key
FK unit_id : int	-> Unit
FK user_id : int	-> UserAccount
relationship_type : varchar	owner tenant
is_primary_payer : boolean	true false
start_date : date	required
end_date : date	NULL = active

BillingPeriod	
Field	Key / rule
PK billing_period_id : int	primary key
period_start : date	UK - one start per cycle
period_end : date	required
due_date : date	required
water_rate_per_cubic_m : decimal	shared period rate
period_status : varchar	draft generated closed
FK created_by : int	-> UserAccount
created_at : datetime	required

MeterReading	
Field	Key / rule
PK meter_reading_id : int	primary key
FK unit_id : int	-> Unit
FK billing_period_id : int	-> BillingPeriod
FK recorded_by : int	-> UserAccount
previous_reading : decimal	meter value
current_reading : decimal	meter value
meter_photo_url : varchar	optional evidence
validation_status : varchar	valid flagged
recorded_at : datetime	required
consumption_cubic_m	CALC = current - previous

UnitBill	
Field	Key / rule
PK unit_bill_id : int	primary key
FK unit_id : int	-> Unit
FK billing_period_id : int	-> BillingPeriod
FK generated_by : int	-> UserAccount
soa_file_url : varchar	generated fixed-layout SOA
soa_generated_at : datetime	required
created_at : datetime	required

BillingForecast	
Field	Key / rule
PK forecast_id : int	primary key
FK unit_id : int	-> Unit
FK forecast_for_period_id : int	-> BillingPeriod
predicted_consumption_cubic_m : decimal	model output
recommendation : text	prescriptive advice
generated_at : datetime	required
predicted_water_charge	CALC from period rate

Notification	
Field	Key / rule
PK notification_id : int	primary key
FK recipient_user_id : int	-> UserAccount
FK unit_bill_id : int	nullable
FK forecast_id : int	nullable
notification_type : varchar	bill payment forecast
channel : varchar	email in_app
message : text	delivery content
delivery_status : varchar	sent failed read
sent_at : datetime	required
read_at : datetime	nullable

BillCharge	
Field	Key / rule
PK bill_charge_id : int	primary key
FK unit_bill_id : int	-> UnitBill
charge_type : varchar	water association_dues
quantity : decimal	water m3 or 1 dues unit
rate_applied : decimal	rate snapshot at billing
description : varchar	optional line description
amount	CALC = quantity x rate_applied

PaymentSubmission	
Field	Key / rule
PK payment_submission_id : int	primary key
FK unit_bill_id : int	-> UnitBill
FK submitted_by : int	-> UserAccount
receipt_url : varchar	uploaded receipt image
ocr_amount : decimal	OCR result
ocr_reference_no : varchar	OCR result
ocr_payment_date : date	OCR result
submitted_at : datetime	required
review_status : varchar	pending approved rejected
FK reviewed_by : int	-> UserAccount; nullable
reviewed_at : datetime	nullable while pending
verified_amount : decimal	approved value
verified_reference_no : varchar	approved value

Why this version is in Third Normal Form (3NF)

- Each relation stores facts about one entity. Every non-key field depends on that table key, not on another non-key field.
- Billing period data is stored once in BillingPeriod, then referenced by MeterReading, UnitBill, and BillingForecast.
- The many-to-many user-to-unit relationship is resolved by UnitAssignment rather than duplicated in UserAccount or Unit.
- No redundant meter_reading_id is stored in UnitBill because both records are uniquely identified by the same unit and billing period.
- Cross-cutting UserAccount, BillingForecast, and Notification FKs are identified in the table rows; they are not all drawn as lines to preserve legibility.

Calculated values are not stored as source data

- Consumption = MeterReading.current_reading - MeterReading.previous_reading.
- Each BillCharge amount = quantity x rate_applied.
- Bill total = SUM(BillCharge.amount) for one UnitBill.
- Amount paid = SUM(PaymentSubmission.verified_amount) where review_status = approved.
- Remaining balance and payment status are API/query-view outputs, not columns in UnitBill.
- Predicted water charge = BillingForecast.predicted_consumption x BillingPeriod.water_rate_per_cubic_m.

3NF constraints to enforce

- UNIQUE(UserAccount.email), UNIQUE(Unit.unit_number), and UNIQUE(BillingPeriod.period_start).
- UNIQUE(UnitAssignment.unit_id, UnitAssignment.user_id, UnitAssignment.start_date).
- UNIQUE(MeterReading.unit_id, MeterReading.billing_period_id).
- UNIQUE(UnitBill.unit_id, UnitBill.billing_period_id).
- UNIQUE(BillCharge.unit_bill_id, BillCharge.charge_type) for the project scope of one water line and one association-dues line per bill.
- Notification should reference one related source: unit_bill_id or forecast_id.